1.

```
from datetime import date


class Student:
    def __init__(self, sid, name, m1, m2, m3, dob_year, dob_month, dob_day, fee_paid):
        self.sid = sid
        self.name = name
        self.m1 = m1
        self.m2 = m2
        self.m3 = m3
        self.dob = date(dob_year, dob_month, dob_day)
        self.fee_paid = fee_paid
        self.total_fee = 50000

    def average_marks(self):
        return (self.m1 + self.m2 + self.m3) / 3

    def cgpa(self):
        return self.average_marks() / 10

    def age(self):
        today = date.today()
        return today.year - self.dob.year

    def fee_balance(self):
        return self.total_fee - self.fee_paid

    def display(self):
        print("Student ID:", self.sid)
```

```
        print("Student Name:", self.name)
        print("Marks:", self.m1, self.m2, self.m3)
        print("Average Marks:", self.average_marks())
        print("CGPA:", self.cgpa())
        print("Age:", self.age())
        print("Fee Balance:", self.fee_balance())
        print("----------------")


class College:
    def __init__(self, code, name, location):
        self.code = code
        self.name = name
        self.location = location
        self.students = []

    def register_student(self, student):
        self.students.append(student)

    def display_details(self):
        print("College Code:", self.code)
        print("College Name:", self.name)
        print("Location:", self.location)
        print("\nStudent Academic Report\n")

        for s in self.students:
            s.display()
```

```
# Creating college
c = College("C101", "ABC Engineering College", "Hyderabad")

# Register 3 students
s1 = Student(1, "Rahul", 85, 90, 80, 2004, 5, 10, 30000)
s2 = Student(2, "Kiran", 75, 88, 82, 2003, 7, 15, 35000)
s3 = Student(3, "Anjali", 92, 89, 95, 2004, 3, 5, 40000)

c.register_student(s1)
c.register_student(s2)
c.register_student(s3)

c.display_details()
```

2.

```
class PersonalInfo:
    def __init__(self, name, dob, phone, email):
        self.name = name
        self.dob = dob
        self.phone = phone
        self.email = email

    def display_personal(self):
        print("Name:", self.name)
        print("DOB:", self.dob)
        print("Phone:", self.phone)
        print("Email:", self.email)
```

```
class Education(PersonalInfo):
    def __init__(self, name, dob, phone, email, degree, university, year, cgpa):
        super().__init__(name, dob, phone, email)
        self.degree = degree
        self.university = university
        self.year = year
        self.cgpa = cgpa

    def display_education(self):
        print("Degree:", self.degree)
        print("University:", self.university)
        print("Year of Graduation:", self.year)
        print("CGPA:", self.cgpa)


class Experience(Education):
    def __init__(self, name, dob, phone, email, degree, university, year, cgpa,
                 company, role, years, skills):
        super().__init__(name, dob, phone, email, degree, university, year, cgpa)
        self.company = company
        self.role = role
        self.years = years
        self.skills = skills

    def display_experience(self):
        print("Company:", self.company)
        print("Role:", self.role)
        print("Years of Experience:", self.years)
```

```
        print("Skills:", self.skills)



class CandidateProfile(Experience):
    def display_profile(self):
        print("\nCandidate Profile")
        self.display_personal()
        self.display_education()
        self.display_experience()

        if self.years > 5:
            print("Eligible for Senior Role")
        else:
            print("Eligible for Junior Role")



# Candidate objects
c1 = CandidateProfile("Ravi", "10-06-1998", "9876543210",
                      "ravi@gmail.com", "B.Tech", "JNTU",
                      2020, 8.2, "TCS", "Software Engineer",
                      6, "Python, Java")

c2 = CandidateProfile("Sneha", "12-08-2000", "9123456789",
                      "sneha@gmail.com", "B.Tech", "JNTU",
                      2022, 8.5, "Infosys", "Developer",
                      2, "Python, SQL")

c1.display_profile()
c2.display_profile()
```

3.class MyMath:

```
    def calculate(self, operation, n):

        if operation == "sum":
            s = n * (n + 1) // 2
            print("Sum of first", n, "natural numbers:", s)

        elif operation == "prime":
            print("First", n, "prime numbers:")
            count = 0
            num = 2
            while count < n:
                for i in range(2, num):
                    if num % i == 0:
                        break
                else:
                    print(num, end=" ")
                    count += 1
                num += 1
            print()

        elif operation == "fibonacci":
            a, b = 0, 1
            print("Fibonacci Series:")
            for i in range(n):
                print(a, end=" ")
                a, b = b, a + b
```

```
        print()

    elif operation == "factorial":
        fact = 1
        for i in range(1, n + 1):
            fact *= i
        print("Factorial of", n, "=", fact)

    else:
        print("Invalid Operation")



# User Input
m = MyMath()

print("1. Sum of Natural Numbers")
print("2. Prime Numbers")
print("3. Fibonacci Series")
print("4. Factorial")

choice = int(input("Enter choice: "))
n = int(input("Enter value of n: "))

if choice == 1:
    m.calculate("sum", n)
elif choice == 2:
    m.calculate("prime", n)
elif choice == 3:
    m.calculate("fibonacci", n)
```

```
elif choice == 4:
    m.calculate("factorial", n)
```